SAR Ship Detection Based on Explainable Evidence Learning Under Intraclass Imbalance

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***Abstract*— Synthetic aperture radar (SAR) ship detection is an important technology supporting water traffic monitoring and marine safety maintenance. In recent years, many methods based on deep neural networks (DNNs) have been used to improve the performance of SAR ship detection. These methods mainly focus on two issues: one is the false alarm of ship detection in complex inshore environments, and the other is the effective extraction and utilization of SAR ship features. The topic discussed in this article is one of the culprits that has caused the aforementioned two problems but has long been overlooked. Specifically, it pertains to the issue of intraclass imbalance in SAR ship detection. There are imbalances in the size distribution, azimuth distribution, and background distribution under the real data collection environment. However, since SAR ship detection is a single-class detection task, the aforementioned imbalances lack reliable descriptors during training. This article proposes using evidence learning to obtain the epistemic uncertainty as a descriptor of biased learning on samples. Contrastive learning is used to further utilize the uncertainty label of samples to correct biased learning under intraclass imbalance. The proposed method is proven to be effective on multiple network models. AP50 reaches 94.8% on the HRSID dataset, 98.4% on the SSDD dataset, and 80.9% on the LS-SSDD dataset, both achieving SOTA performance.**

***Index Terms*— Contrastive learning, evidence learning, intraclass imbalance, synthetic aperture radar (SAR) ship detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) has been widely used in the field of remote-sensing image interpretation due to its all-weather observation capability [1], [2]. Among them, SAR ship detection supports water traffic monitoring and marine security maintenance. In recent years, a large number of successful applications of AI technology have emerged in the field of remote sensing. The deep-learning methods represented by a convolutional neural network (CNN) help researchers better model and fuse the spatial features [3], [4], [5] and temporal features [6] of remote-sensing images. SAR ship detection also benefits from these studies. Most of these studies focus on two key issues: reducing false alarms in inshore ship detection and effectively extracting and utilizing SAR ship features. Researchers have explained and solved these two issues from different perspectives. Du et al. [7] added scene branch intelligence to distinguish inshore and offshore scenarios and designed different detection strategies based on the scenarios to reduce false alarms in inshore detection. Zhou et al. [8] utilized a dual pooling reconstruction feature extraction module to reduce sidelobe effects. Xiong et al. [9], [10] focused on SAR weak target detection by obtaining feature images through feature transformation weakly correlated with SNR.

The emergence of the above two problems is not only related to the imaging characteristics of SAR images themselves, but also to the particularity of ship detection tasks. First, ship datasets are collected in real environments, so there is often a problem of imbalanced samples. This imbalance is within the class because ship detection is a single-class detection task. Second, SAR images themselves are affected by noise and are sensitive to azimuth angles, also resulting in sample imbalances. These imbalances are summarized as follows.

1) The types and sizes of ships are imbalanced.
2) The samples of ships from different azimuth angles are imbalanced.
3) The impact of noise on the ship samples is imbalanced.
4) The environmental distribution of the ship samples is imbalanced.

The above four types of intraclass imbalance lead to incomplete feature extraction of targets and false alarms in the detection of inshore ships due to the biased learning of the model. It can be seen that the difficulty of the intraclass imbalance problem is that there are no explicit or implicit labels to distinguish samples with poor model fitting. We refer to the samples with poor model fitting as hard samples in training. Generally speaking, the minority samples in the imbalance are hard. But at the same time, the majority samples with complex or weak characteristics may also be hard. The two problems of incomplete feature extraction of targets and inshore false alarms are unified as the hard sample problem under intraclass imbalance based on this perspective. Epistemic uncertainty is introduced to label hard samples under intraclass imbalance in this article.

We have used deep neural networks (DNNs) to automatically extract features from SAR images. We trust the normalized probability output by the model as the

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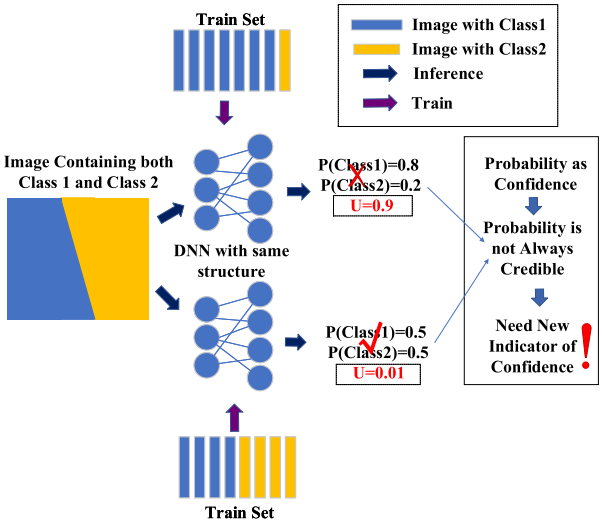


Fig. 1. Unreliable output of DNN model driven by data (P: output of the model, normalized as probability and U: uncertainty of model output, which is a reliable indicator of confidence of model output).

classification confidence all along. However, the resulting predictions lack reliable explanations. Fig. 1 shows an example of data-driven model inference influenced by the training set. The output of the DNN model is controlled by the distribution of the training dataset for an image with multiclass. When the model outputs $p(\text{class1}) = 0.8$ and $p(\text{class2}) = 0.2$, it gives a high-confidence false prediction that the image is class1. The DNN model needs tools to provide reliable explanations for the prediction. For the model output with $p(\text{class1}) = 0.5$ and $p(\text{class2}) = 0.5$, if only the normalized classification probability is considered, the model might be deemed as making uncertain judgments by ordinary users. However, this prediction happened to be proof of the model's reliability and robustness.

We recognize that the so-called hard samples are those that are not fully learned and confidently recognized by the model. The reliability explanation of model output is the key to solving the problem of hard samples under intraclass imbalance. Epistemic uncertainty is used to provide interpretable sample labels in training. Epistemic uncertainty quantifies how robust the model is to modeling the current sample. The model tends to be more stable and biased toward the majority class samples [11] in the case of imbalance, resulting in insufficient feature learning for the minority class samples [12]. It is manifested in the insufficient learning of ship features with rare models and a small number of special imaging angles, resulting in high epistemic uncertainty in ship detection. Huang et al.'s [13] research on explainable uncertainty in ship detection confirms this point.

There are other studies of model explainable uncertainty in other areas of SAR image interpretation. For example, [14], [15], and [16] introduced uncertainty analysis into ice–water classification. However, it is difficult to transfer these studies to the field of object detection and further optimize object detection models using interpretable uncertainty. The main reason is that most of the above studies are based on variational inference methods for Bayesian networks. Bayesian neural networks obtain epistemic uncertainty by learning the posterior of model weights rather than fixed weights. In other words, variational inference obtains approximate solutions by sampling weights. Implementing variational inference is intricate and challenging to apply across diverse object detection frameworks. Consequently, Huang et al. [13] used the MC dropout method in SAR ship detection, which is a special case and simple implementation of variational inference. Specifically, a dropout layer is added to the network to increase random perturbation and weights are randomly sampled multiple times to calculate epistemic uncertainty. There are two main problems with this approach. First, multiple sampling increases the training time of the model. Second, the introduction of dropout can damage the performance of the model. This issue is uncontrollable in object detection because the model is sensitive to the location and dropout probability of the dropout layer. The disadvantage of MC dropout makes it difficult to directly use the uncertainty obtained from it to optimize the target detection model.

This article chooses evidence learning [17] to obtain uncertainty. Evidence learning interprets the standard softmax output of a classification network as a parameter set of class distributions. By replacing this parameter set with the parameters of Dirichlet density, the model's prediction is represented as a distribution of softmax outputs, rather than a point estimate of the softmax output. The uncertainty is directly calculated from the model's prediction. More importantly, evidence learning does not require multiple sampling training. This article also proves that evidence learning does not affect the performance of the target detection model.

Upon acquiring interpretable uncertainty labels, this article introduces contrastive learning to relieve the intraclass imbalance in model learning. Previous studies on the intraclass imbalance of image recognition still use classification probability as an indicator of intraclass imbalance [18], [19], which is not accurate enough. Meanwhile, these studies utilize indirect methods such as repeated sampling and data enhancement to optimize model performance. These methods are hardly transferred to SAR detection tasks because target-level data augmentation is difficult to achieve. They also have poor training efficiency. Compared to them, our approach is more direct and efficient. Our specific method is to divide samples into low uncertainty and high uncertainty according to the threshold. Quantitative uncertainty can be used as a basis for detecting difficult samples in the model. Specifically in ship detection, the uncertainty of false alarm targets that are easily confused near the coast is low (features similar to ships). The uncertainty of minority ship targets in intraclass imbalance is high, such as rare types of ships and those with severe geometric distortions. In the positive samples, the low-uncertainty samples are compared with the low-uncertainty and high-uncertainty samples, respectively. The comparison between low-uncertainty samples is to further learn and enrich the sample features. The comparison between low-uncertainty samples and high-uncertainty samples is to shorten the distance between the outlier-hard sample features and other features. This is mainly to further utilize the

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features of SAR ships and solve the biased learning caused by intraclass imbalance. Simultaneously, negative samples are compared with low-uncertainty positive samples. The distance between the features of the two types of samples is pushed away. Negative samples with features similar to ships have low uncertainty, and the decision curve between such negative samples and positive samples should be clear. At this point, the two problems of inshore false alarms and incomplete feature extraction caused by intraclass imbalance have been solved from a unified perspective.

The main contributions of this article are as follows.

1) Based on our knowledge, this is the first time that model interpretability has been used to solve the data imbalance problem in the field of SAR image interpretation. This article solves the intraclass imbalance problem in SAR ship detection from the perspective of model interpretability, and the degree of intraclass imbalance is properly judged by the concept of epistemic uncertainty.
2) We introduce evidence learning into the model interpretable analysis of SAR automatic target recognition (ATR) for the first time. Compared to other uncertainty analysis methods, evidence learning has significant advantages in terms of architecture flexibility, training efficiency, and model accuracy. These advantages make it possible for us to optimize the performance of the ship detection model using uncertainty, rather than focusing solely on model robustness like previous research on uncertainty in the SAR field.
3) With the help of uncertainty measuring the degree of intraclass imbalance, a method relying on contrastive feature constraints to alleviate intraclass imbalance is proposed. The method optimizes model performance directly during training and can be naturally migrated to many existing SAR ship detection networks. It achieves SOTA results on the HRSID, SSDD, and LS-SSDD datasets.

II. Related Work

A. SAR Ship Detection

SAR ship detection is an area that has long received widespread attention. With the advancement of deep-learning technology, many detection methods and datasets based on deep learning have been developed. Two-stage networks such as Faster-RCNN [20] and single-stage networks such as YOLO [21] have been introduced into SAR ship detection and a lot of adaptive optimization was done. Due to different imaging conditions, popular SAR ship datasets such as SSDD [22], HRSID [23], and LS-SSDD [24] exhibit differences in background environment, resolution, and imaging angle, resulting in an imbalanced limited knowledge phenomenon. For example, in the HRSID dataset, large ships only account for 2% of the samples, and offshore scene images account for 18.4%. Detection models trained on these datasets cannot extract complete features and are prone to generate false alarms in inshore scenarios. To address these two problems, researchers have provided a variety of solutions. In terms of improving feature extraction, Jiang et al. [25] used the CBAM-DLA34 backbone to finely extract ship features. Zhu et al. [26] redesigned the parameter distribution in the network and improved the multiscale detection capability of ships through feature reuse. In terms of false alarm suppression, Zhang et al. [27] proposed a multiscale inference fusion (MSIF) framework that integrates two methods of eliminating false alarms, namely pixel feature filtering and multisource fusion, to eliminate false alarms in large-scale SAR images and improve the overall performance of ship detection. Zhou et al. [8] proposed a sidelobe sensing module to reduce the interference of sidelobes, using double pooling to reduce the influence of sidelobes, and proposed a new coIOU loss to reduce false positives (FPs) and false negatives of ships.

B. Epistemic Uncertainty in the DNN and Its Application in SAR

Two types of uncertainty in DNNs were originally defined by Kendall and Gal [28]. They divided uncertainty into epistemic uncertainty, which explains the uncertainty in model weights, and aleatoric uncertainty, which captures the uncertainty in data noise. For epistemic uncertainty, Bayesian neural networks are used for modeling. Bayesian neural networks treat each weight as a distribution rather than a fixed value. Variational inference [29], Hamiltonian Monte Carlo [30], and other methods are used to approximate the solution of Bayesian inference. Gal and Ghahramani [31] have verified that Monte Carlo dropout is a special case and simple implementation of variational inference, making it easy to embed DNNs into various network structures. As follows, $p(w|D)$ is the posterior distribution of the model weight w under the condition of training data $D = (X, Y)$. The prediction distribution of Bayesian neural networks does not use a fixed set of weights, but instead uses a posterior probability weighting of all parameters

$$p(y*|x*, D) = \int p(y*|x*, w)p(w|D)dw \tag{1}$$

where $x*$ is the test image and $y*$ represent the prediction results. In practice, the integral in (1) is achieved by MC approximation

$$p\left(y^*|x^*, D\right) \approx (1/N)\sum_{i=1}^{N} p\left(y^*|x^*, w_i\right)p(w_i|D) \tag{2}$$

where the sampling from the posterior $p(w_i|D)$ is also approximated. MC dropout uses dropout to randomly discard neurons during model training to simulate sampling and achieve the purpose of approximating $p(w|D)$.

Epistemic uncertainty indicates the safety and reliability of deep model predictions and has received particular attention in some computer vision tasks, such as medical diagnosis [32] and autonomous driving [33]. In the field of SAR image interpretation, water–ice classification first introduced uncertainty verification to validate the reliability of model predictions [14], [15], and it is expected to improve model performance with epistemic uncertainty [16]. For SAR target detection, knowledge of whether the model prediction is reliable is necessary, because it is difficult for untrained ordinary users to distinguish SAR targets like optical remote-sensing

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targets. Therefore, Huang et al. [13] conducted a detailed study of epistemic uncertainty in SAR target detection. However, there are obstacles to using uncertainty to optimize SAR target detection. The popular MC dropout method sometimes compromises the detection performance of the model.

Evidence learning thus introduces the study of epistemic uncertainty. Instead of sampling to obtain approximate Bayesian solutions, evidence learning introduces subjective logic to measure the uncertainty of the model. Sensoy et al. [17] first proposed a method for calculating the uncertainty of evidence in image classification tasks. Amini et al. [34] calculated the uncertainty of evidence in continuous regression tasks. This article will use evidence learning to calculate epistemic uncertainty in the classification head of object detection models.

III. Methodology

As shown in Fig. 2, the proposed method is a two-stage learning model. During the first stage, an evidence-learning head is added to a conventional object detection model. In addition to outputting target classification and location information, the model outputs epistemic uncertainty through the evidence-learning head. The positive and negative samples predicted by the model are classified into low and high uncertainty according to a threshold value after complete rounds of learning. In the second stage, uncertainty is injected as an additional label into the training data labels. Initializing with the best-performing model from the first stage, we perform further fine-tuning for a reduced number of epochs. In contrast to the first stage, a loss function based on three types of contrastive learning using uncertainty is added to the training process within the initial five epochs.

A. Modeling Epistemic Uncertainty Based on Deep Evidence Learning

In SAR ship detection, the cost of trust mispredictions can be reduced by predicting epistemic uncertainty. More importantly, epistemic uncertainty can serve as additional knowledge to further improve model performance. However, the popular MC dropout method has the problem of high computational cost and damaging model performance. Therefore, we introduced subjective logic [35] to consider the epistemic uncertainty of the model.

For conventional neural networks, the output probability p_i for any sample i is provided by the likelihood function, which is determined by the model's weight parameters. In the Bayesian perspective, for probability p_i, we use the Dirichlet distribution $D(p_i|\alpha_i)$ to measure the probability density distribution (second-order probability). The Dirichlet distribution is controlled by K parameters, where K is equal to the number of categories

$$D(p|\alpha) = \begin{cases} \dfrac{1}{B(\alpha)} \displaystyle\prod_{i=1}^{K} p_i^{\alpha_i - 1} & \text{for } p \in S_k \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

where $B()$ is the beta function, and $S_K = \{p | \sum_{k=1}^{K} p_k = 1, 0 \le p_k \le 1, \forall k\}$. It should be noted that for SAR ship detection that only distinguishes background and targets, the Dirichlet distribution will be degenerated into the Beta distribution when $K = 2$. However, for a more general description of the working process of the evidence head, the following discussion does not assume $K = 2$. Using Dempster–Shafer Evidence Theory (DST), for sample i, each label's probability p_i was assigned a belief mass, denoted as b_k

$$u + \sum_{k=1}^{K} b_k = 1 \tag{4}$$

where u is uncertainty. The belief mass is obtained through evidence. In DST, evidence is the measure of data supporting samples falling into a certain class. In this article, the evidence for each class $e = [e_1, e_2, \ldots, e_k]$ is directly obtained from the output of the neural network. Then, the parameters of Dirichlet distribution in (3) can be computed by using $\alpha_k = e_k + 1$. In fact, $\alpha_k = e_k + \beta$, β represents prior information, and taking 1 represents no prior. Define Dirichlet strength S

$$S = \sum_{k=1}^{K} \alpha_k = \sum_{k=1}^{K} (e_k + 1). \tag{5}$$

The uncertainty and belief masses are determined by the parameters as

$$u = \frac{K}{S} \tag{6}$$

$$b_k = \frac{\alpha_k - 1}{S}. \tag{7}$$

The advantage of calculating uncertainty this way is that it avoids considering the probability of the labeled class alone as uncertainty and takes into account the comparison values with other classes [17]. By combining (4), (6), and (7), it can be seen that when the evidence for all categories is insufficient to support predictions, the belief mass assigned to each category will also be low. At this time, the epistemic uncertainty of the output is high, indicating a high likelihood of prediction errors, that is, unreliable predictions.

To ensure that the evidence e_i is positive, the softmax layer of the evidence head is replaced with a nonnegative activation function. The exp function or relu function is used based on the range of logits values l_i output by the network. Both activation methods are only used to ensure nonnegative output and there is no significant difference in performance

$$e_i = \exp(l_i) \tag{8}$$

$$e_i = \text{relu}(l_i). \tag{9}$$

The evidence head performs a classification task with a prior, which is still used to distinguish ship targets from the background. We use Type II Maximum Likelihood (Empirical Bayes) [36] to set optimization goals. First, we obtain the evidence e_i from the evidence head and convert the class label into a one-hot vector y_i. Second, we treat an adjusted Dirichlet distribution $\widetilde{D}(p_i|e_i)$ as the prior of multinomial likelihood $P(y_i|p_i)$ (the classification opinion). The loss function is the negative logarithm of marginal likelihood

$$L = -\log\left[\int \prod_{k=1}^{K} p_{ik}^{y_{ik}} \frac{1}{B(e_i)} \prod_{k=1}^{K} p_{ik}^{e_{ik}-1} d\,p_i\right]$$

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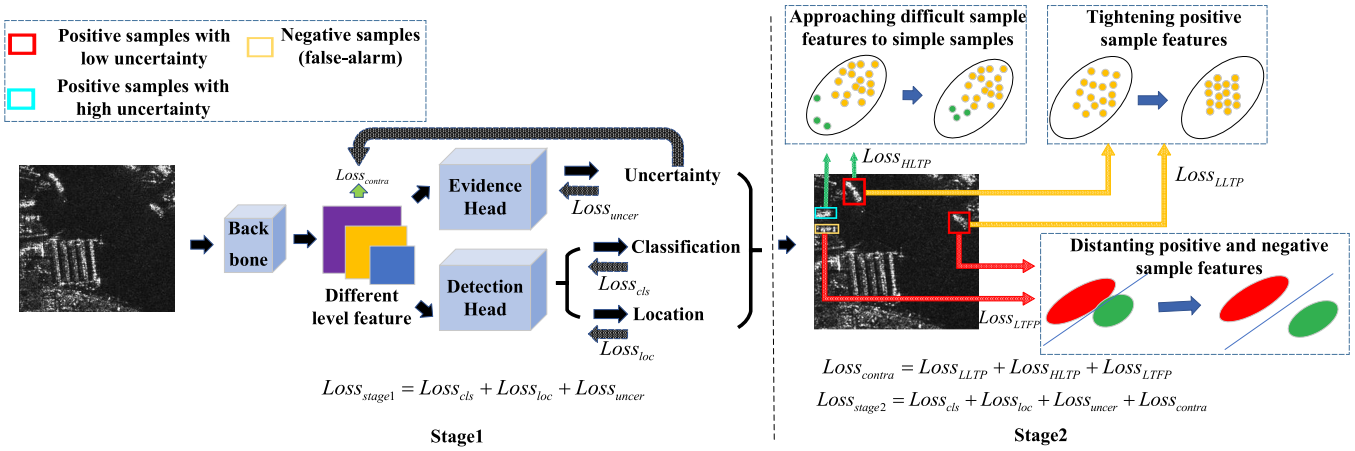


Fig. 2. Training process is divided into two stages. Stage 1: Evidence head is added to the detection model. Uncertainty is calculated through evidence. Stage 2: The prediction results of the first stage are divided into three types of samples based on uncertainty. Using these samples, three types of contrastive learning are performed on the multilevel feature maps of the backbone).

$$= \sum_{k=1}^{K} y_{ik} \left[\log\left(\sum_{k=1}^{K} e_{ik} \right) - \log(e_{ik}) \right]. \tag{10}$$

The loss function proposed in (10) has certain limitations. The current goal is to ensure that the correct class is assigned more evidence compared to other classes, but the overall evidence of the error class should also be low. To avoid the uncertainty being too low due to the excessively high overall evidence, the KL divergence-based correction loss function [17] is introduced

$$\begin{aligned} \mathcal{L}_{kl} &= \mathrm{KL}\left(D\left(\boldsymbol{p}_i \mid \tilde{\boldsymbol{\alpha}}_i\right) \| D\left(\boldsymbol{p}_i \mid \mathbf{1}\right)\right) \\ &= \log \frac{\Gamma\left(\sum_{k=1}^{K} \tilde{\alpha}_k\right)}{\Gamma(K) \prod_{k=1}^{K} \Gamma\left(\tilde{\alpha}_{ik}\right)} \\ &\quad + \sum_{k=1}^{K} \left(\tilde{\alpha}_{ik} - 1\right) \left[\psi\left(\tilde{\alpha}_{ik}\right) - \psi\left(\sum_{k=1}^{K} \tilde{\alpha}_k \right) \right] \end{aligned} \tag{11}$$

where $\tilde{\alpha}_i = 1 + (1 - y_i) \odot e_i$ is the adjusted Dirichlet parameters, $\Gamma()$ is the gamma function, and $\psi()$ is the digamma function. Uncertainty is avoided from being unreasonably low by minimizing the distance between the Dirichlet distribution adjusted by (10) and the target distribution. Finally, the loss of evidence head is

$$L_{\text{evi}} = L + \lambda L_{\text{kl}} \tag{12}$$

where λ is a hyperparameter related to the epoch of training

$$\lambda = \min\{1, t/T\} \tag{13}$$

where T is the total number of epochs and t is the current epoch. KL divergence gradually intervenes in the loss function as training progresses, preventing the evidence learned by the model from being too flat at the beginning.

B. Contrastive Learning Based on Uncertainty

As shown in the training process in Fig. 2, after the target detection model with the evidence head has undergone a complete number of epochs of training, the predicted ship position and epistemic uncertainty are obtained. As an indicator of the reliability of the reaction model's prediction, the epistemic uncertainty also reflects the degree of learning by the model on the current sample. So, after obtaining epistemic uncertainty, we divide the prediction results into high uncertainty and low uncertainty between true positive (TP, the total number of ships correctly detected) and false positive (FP, the total number of ships incorrectly detected) through thresholds.

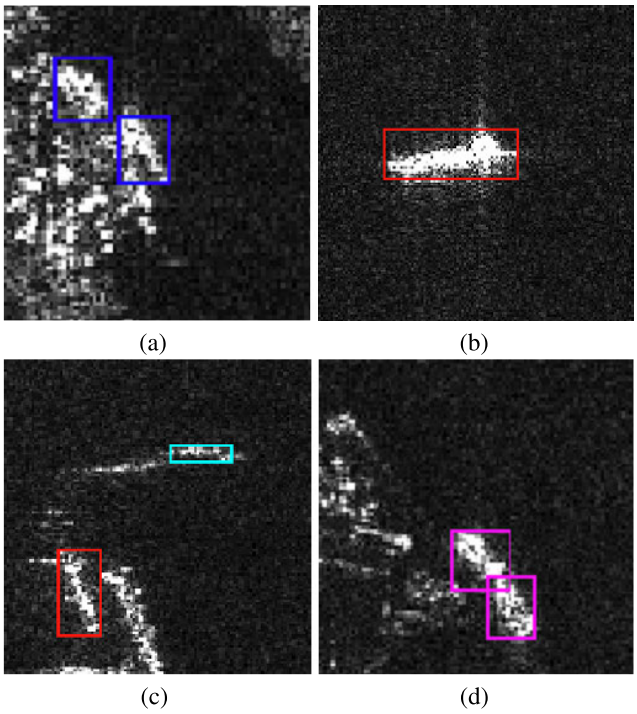


Fig. 3. Visualization of ship uncertainty (HTP: dark blue; LTP: red; HFP: light blue; and LFP: magenta). (a) HTP. (b) LTP. (c) HFP. (d) LFP.

As shown in the upper part of Fig. 3, the model has low uncertainty and high confidence for SAR ships in open ocean environments, and high uncertainty for ship samples in complex inshore environments. On the one hand, the HRSID ship dataset accounts for 18.4% of inshore scene images, and the training frequency of inshore ship samples is low; On the other hand, the inshore environment is complex, making it difficult to extract features from ship samples. The uncertainty obtained

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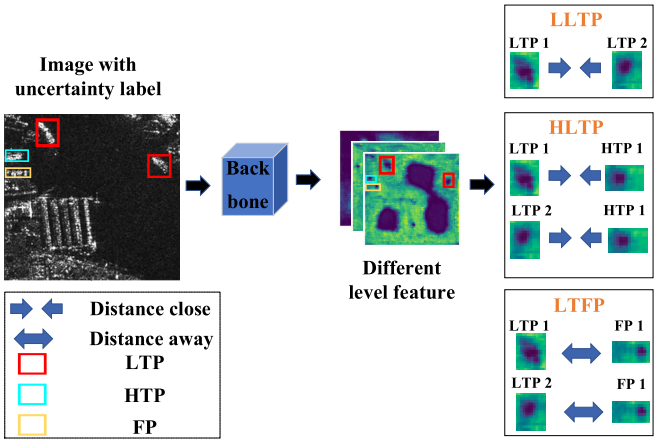


Fig. 4. Three groups of contrastive learning on the feature map.

from the evidence head well expresses the robustness of the model prediction, which is consistent with our understanding of the model training process. The lower part of Fig. 3 shows the error prediction results of the model. The HFP sample indicates that although the model gives wrong predictions, it has a high degree of uncertainty. LFP samples indicate that the model is very confident in its wrong predictions. It can be seen that HFP is an obvious false alarm, while LFP is easily confused with high uncertainty TP (HTP).

It is a natural thing to further optimize the model performance using the uncertainty. Based on the qualitative experiment shown in Fig. 3, a model performance improvement module based on contrastive learning that utilizes uncertainty is proposed. First, the uncertainties and positions of HTP, LTP, HFP, and LFP are injected into the training data as additional labels. Then, these additional sample labels were divided into three groups.

1) *HLTP:* HTP and TP with low uncertainty (LTP).
2) *LLTP:* LTP and LTP.
3) *LTFP:* LTP and FP.

As shown in Fig. 4, modern DNN-based target detection networks use multilevel feature maps extracted by the backbone network for detection. HTP, LTP, HFP, and FP are grouped and directly used in training to extract target features based on location information on multilevel feature maps for contrastive learning. Within each group, contrastive learning using cos distance is added to the loss function. It should be noted that although LFP is more easily confused with TP, we still choose to use all FP for contrastive learning. The comparison between HFP and LTP is still valuable.

For the first group (HTP and LTP), assuming that the target detection network has N layers of feature maps, for feature map F_i, the feature obtained by the jth HTP label according to its position is f_{htp}^j and the feature obtained by LTP is f_{ltp}^j. The loss function of the ith layer is as follows:

$$L_{\text{HLTP}}^i = \sum_j \left(1 - \cos < f_{\text{htp}}^j, f_{\text{ltp}}^j >\right) = 1 - \frac{f_{\text{htp}}^j \cdot f_{\text{ltp}}^j}{||f_{\text{htp}}^j|| ||f_{\text{ltp}}^j||}. \tag{14}$$

The overall loss function is the weighted sum of each layer

$$L_{\text{HLTP}} = \sum_{i=1}^{N} \lambda^i L_{\text{HLTP}}^i. \tag{15}$$

The setting of weight parameters λ can refer to the layer balance hyperparameters of target detection. Some target detection frameworks set the weights of each layer participating in learning for different tasks. If not, they are set to 1. From (14), it can be seen that the setting of the first set of contrastive learning is to narrow the distance between HTP and LTP, that is, to make the features of outlier high uncertainty samples close to the low-uncertainty samples that the model has learned well.

For the second group (LTP and LTP), the loss function based on cos distance is still used

$$L_{\text{LLTP}}^i = \sum_j \left(1 - \cos < f_{\text{ltp}}^j, f_{\text{ltp}}^j >\right) = 1 - \frac{f_{\text{ltp}}^j \cdot f_{\text{ltp}}^j}{||f_{\text{ltp}}^j|| ||f_{\text{ltp}}^j||}. \tag{16}$$

The overall loss function is the weighted sum of each layer

$$L_{\text{LLTP}} = \sum_{i=1}^{N} \lambda^i L_{\text{LLTP}}^i. \tag{17}$$

The distance between two LTP sample features being shortened can make the within-class of TP more compact because LTP samples account for the majority of TP samples.

The comparison of the last group (LTP and FP) mainly serves to reduce the ship's inshore false alarm. As previously discussed, LFP represents the worst case scenario for model prediction, where the model gives a high confidence incorrect prediction. Confusion between LFP and TP is easily incurred, and the classification boundary of the model is vague and unreliable between them. HFP is a more obvious false alarm, but further differentiation from TP is still needed. The comparison between LTP and FP is to push the distance between their features. It should be noted that although confusion between HTP and FP is more likely to occur (HTP is the correct prediction with low confidence), LTP is still chosen for comparison. On the one hand, LTP has more stable features compared to HTP because the model learns better. On the other hand, there are more LTP samples, and the model can learn more abundant contrastive knowledge. The loss function for the third group is as follows:

$$L_{\text{LTFP}}^i = \sum_j \cos < f_{\text{ltp}}^j, f_{\text{fp}}^j > = \frac{f_{\text{ltp}}^j \cdot f_{\text{fp}}^j}{||f_{\text{ltp}}^j|| ||f_{\text{fp}}^j||}. \tag{18}$$

The overall loss function is the weighted sum of each layer

$$L_{\text{LTFP}} = \sum_{i=1}^{N} \lambda^i L_{\text{LTFP}}^i. \tag{19}$$

After obtaining three sets of comparative losses, the overall contrastive losses were obtained

$$L_{\text{contra}} = L_{\text{HLTP}} + L_{\text{LLTP}} + L_{\text{LTFP}}. \tag{20}$$

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The last thing worth noting is the training process of contrastive learning. First, the training of contrastive learning intervention is a short number of epochs of fine-tuning, usually within ten epochs. Second, during the fine-tuning process, contrastive learning loss is added only in the first five epochs. This is considered for the following two main points.

1) As a two-stage learning method, the uncertainty label obtained in the first stage is based on the target detection model of complete epoch learning, which is relatively stable. However, contrastive learning based on these uncertainty labels should still not be done too much. It can be seen that the contrastive learning part of the proposed method can be performed iteratively.
2) The contrastive loss acts directly on the backbone of the object detection model. The detection model is sensitive to such adjustments, and fine-tuning after removing the contrast loss can make the model converge quickly.

IV. Experiments and Discussion

To fully validate the proposed method, this article uses three publicly available ship datasets to verify the detection performance of this article: HRSID [23], LS-SSDD [37], and SSDD [38]. The proposed method will be embedded into two advanced object detection models, namely RTMDET [39] and sidelobe-aware YOLOv5(SA YOLOv5) [8].

The experiment will be divided into two parts. The first part of the ablation experiment aims to verify the advantages of evidence learning compared to MC dropout and the effectiveness of comparative learning. The second part is a comparative experiment to demonstrate the superiority of the method.

A. Experimental Data and Baseline Model

HRSID and LS-SSDD are both popular large SAR ship datasets that include large ship targets, small ship targets, inshore ship targets, and offshore ship targets. Their information is shown in Table I.

1) The HRSID dataset is collected from Sentinel-1 and TerraSAR-X, for ship instance segmentation including ship detection. There are 5604 SAR images with 0.5-, 1-, and 3-m resolutions; 800 × 800 pixels. 16 951 ships are contained in HRSID.
2) The LS-SSDD dataset images are taken by Sentinel-1, which consists of 15 images with a size of 24 000 × 16 000 pixels. The large-scale image is cut into 9000 subimages with a size of 800 × 800 pixels in this article. The image resolution is 5–20 m. There are 6015 ships in LS-SSDD-v1.0.
3) The SSDD dataset images are collected from RADARSAT-2, TerraSAR-X, and Sentinel-1 sensors. The dataset contains 1160 SAR images of various image sizes, such as 416 × 323 and 501 × 355. The resolution ranges from 1 to 15 m. There are 2551 ships in total in SSDD. Among them, there are 1463 small ships, 989 medium ships, and 99 large ships. A total of 928 images are used as a training set, and 232 images are used as a testing set in this article.

The two baseline models selected in this article perform well on remote-sensing datasets. RTMDET is an advanced object detection model developed based on YOLOX [40], which achieves SOTA performance on the optical remote-sensing dataset DOTA [41]. SA YOLOv5 is based on YOLOv5 [42] and has undergone multiple optimizations for SAR ship detection tasks, including sidelobe cancellation modules and more reasonable regression loss functions, achieving SOTA performance on ship detection datasets such as LS-SSDD.

B. Model Hyperparameters and Environment Settings

The model training hyperparameters in this article follow the same empirical settings. For the first stage of training of the proposed method, both RTMDET and SA YOLOv5 are trained using the stochastic gradient descent (SGD) approach, with the learning rate set at 0.001. Momentum is set to 0.937, and the weight decay is set to 5e-4. The maximum number of training epochs is set to 100. The learning rate is reduced to 0.0001 at the 50th epoch to facilitate the search for the optimal model. The initial model in the second stage is the best-performing model in the first stage, so the learning rate is still 0.0001 and trained for 20 epochs.

The software environment used in experiments is unified as Pytorch-1.13.1, CUDA 11.6, and cuDNN 8.2.1. The hardware environment is given as follows: the CPU is AMD EPYC 7501, and the GPU is GeForce RTX 3090.

C. Evaluation Metrics

Precision (P), recall (R), $F1$ Score, and average precision (AP) are used to evaluate the performance of the detection model. The precision is defined as the proportion of correctly detected ships among all detected ships

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{21}$$

Recall indicates the fraction of detected ships relative to all-round truth, denoted as

$$\text{recall} = \frac{TP}{\text{TP} + \text{FN}} \tag{22}$$

where FN represents false negatives, also known as missed detection.

The $F1$ Score is determined by recall and precision and is defined as

$$F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{23}$$

AP is also determined by recall and precision and is defined as

$$\text{AP} = \int_0^1 P(R) dR. \tag{24}$$

$P(R)$ is expressed as the curve of precision and recall, which can be interpreted as the area contained by the $P(R)$ curve and the coordinate axis. The $F1$ Score reflects the single-point performance of the model on the $P(R)$ curve. Compared with the $F1$ Score, AP can more comprehensively reflect the performance of the target detection model. AP is affected by

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TABLE I
DETAILS OF THE DATASET

Parameter	HRSID	LS-SSDD	SSDD
Satellite	Sentinel-1, TerraSAR-X	Sentinel-1	RadarSat-2, Sentinel-1, TerraSAR-X
Resolution	0.5m $\sim$ 3m	5m $\sim$ 20m	1m $\sim$ 15m
Image size	800 $\times$ 800	800 $\times$ 800	416 $\times$ 323, 501 $\times$ 355, etc
Image Number	5604	9000	1160
Object Number	16951	6015	2551

TABLE II
RTMDET PERFORMANCE ON HRSID

Model	P(%)	R(%)	F1	AP50(%)	AP75(%)	AP(0.5:0.95)(%)	FPS
Baseline	57.3	91.6	70.5	89.6	78.2	69.3	**36.0**
MC dropout	57.4	91.5	71.0	89.4	77.3	68.0	36.0
EDH	57.2 (-0.1)	93.0 (+1.4)	70.8 (+0.3)	91.1 (+1.5)	81.3 (+3.1)	71.6 (+2.3)	35.2 (-0.8)
EDHC	**66.2 (+8.9)**	**92.9 (+1.3)**	**77.3 (+6.8)**	**91.2 (+1.6)**	**81.8 (+3.6)**	**72.8 (+3.5)**	35.2 (-0.8)

TABLE III
SA-YOLOV5 PERFORMANCE ON HRSID

Model	P(%)	R(%)	F1	AP50(%)	AP75(%)	AP(0.5:0.95)(%)	FPS
Baseline	93.9	**90.0**	91.9	94.0	80.9	70.4	**37.7**
MC dropout	93.2	89.9	91.5	93.9	79.8	69.7	37.7
EDH	93.8 (-0.1)	89.7 (-0.3)	91.7 (-0.2)	94.5 (+0.5)	82.0 (+1.1)	71.4 (+1.0)	37.3 (-0.4)
EDHC	**94.8 (+0.9)**	89.2 (-0.8)	**91.9 (+0.0)**	**94.8 (+0.8)**	**84.0 (+3.1)**	**72.6 (+2.2)**	37.3 (-0.4)

TABLE IV
RTMDET PERFORMANCE ON LS-SSDD

Model	P(%)	R(%)	F1	AP50(%)	AP75(%)	AP(0.5:0.95)(%)	FPS
Baseline	43.2	76.2	55.1	68.2	7.6	23.1	**36.0**
MC dropout	37.9	77.5	50.9	69.7	7.5	23.8	36.0
EDH	47.4 (+4.2)	77.8 (+1.6)	58.9 (+3.8)	70.3 (+2.1)	7.6 (+0.0)	23.3 (+0.2)	35.2 (-0.8)
EDHC	**50.2 (+7.0)**	**77.8 (+1.6)**	**61.0 (+5.9)**	**70.6 (+2.4)**	**9.1 (+1.5)**	**23.8 (+0.7)**	35.2 (-0.8)

TABLE V
SA-YOLOV5 PERFORMANCE ON LS-SSDD

Model	P(%)	R(%)	F1	AP50(%)	AP75(%)	AP(0.5:0.95)(%)	FPS
Baseline	83.1	70.5	76.3	77.1	14.0	29.4	**37.7**
MC dropout	81.8	72.6	76.9	77.9	14.1	29.7	37.7
EDH	85.9 (+2.8)	72.1 (+1.6)	78.4 (+2.1)	79.5 (+2.4)	15.3 (+1.3)	31 (+1.6)	37.3 (-0.4)
EDHC	**85.9 (+2.8)**	**72.9 (+2.4)**	**78.9 (+2.6)**	**80.9 (+3.8)**	**16.0 (+2.0)**	**32.2 (+2.8)**	37.3 (-0.4)

the IOU threshold. This article follows the AP calculation method of the COCO dataset, calculating AP50, AP75, and AP (0.5:0.95), respectively. AP50 is the AP under the IOU score of 0.5 of the detection box.

1) AP50 is the AP under the IOU score of 0.5 of the detection box.
2) AP75 is the AP under the IOU score of 0.75 of the detection box. AP75 is a more stringent performance metric than AP50.
3) AP (0.5:0.95) represents the average AP at different IoU thresholds (0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95).

D. Performance and Analysis of the Proposed Method

To verify the effectiveness and rationality of the proposed method, the following experimental groups were set up.

1) Normal trained object detection model (Baseline).
2) Object detection model using MC dropout to obtain uncertainty (MC dropout).
3) Object detection model with added evidence head to obtain uncertainty (EDH).
4) Object detection model with added evidence head and contrastive learning (EDHC).

The experimental results on the HRSID dataset are shown in Tables II and III.

The experimental results on the LS-SSDD dataset are shown in Tables IV and V.

The experimental results on the SSDD dataset are shown in Tables VI and VII.

As can be seen from Tables II–VII, the impact of the MC dropout method calculation uncertainty on model detection performance is uncontrollable and may cause negative impacts

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TABLE VI
RTMDET PERFORMANCE ON SSDD

Model	P(%)	R(%)	F1	AP50(%)	AP75(%)	AP(0.5:0.95)(%)	FPS
Baseline	48.5	98.9	65.1	97.7	87.0	71.0	**36.0**
MC dropout	50.7	96.9	66.6	96.3	79.9	67.0	36.0
EDH	53.0 (+4.5)	98.9 (+0.0)	69.0 (+3.9)	97.8 (+0.1)	87.1 (+0.1)	70.7 (-0.3)	35.2 (-0.8)
EDHC	**70.0 (+21.5)**	**99.1 (+0.2)**	**82.0 (+16.9)**	**98.4 (+0.7)**	**89.0 (+2.0)**	**71.1 (+0.1)**	35.2 (-0.8)

TABLE VII
SA-YOLOV5 PERFORMANCE ON SSDD

Model	P(%)	R(%)	F1	AP50(%)	AP75(%)	AP(0.5:0.95)(%)	FPS
Baseline	96.5	97.1	96.8	98.5	82.4	70.5	**37.7**
MC dropout	96.9	96.3	96.6	98.1	80.7	70.1	37.7
EDH	96.5 (+0.0)	96.2 (-0.9)	96.3 (-0.5)	98.4 (-0.1)	**83.3 (+0.9)**	70.5 (+0.0)	37.3 (-0.4)
EDHC	**96.7 (+0.2)**	**97.4 (+0.3)**	**97.1 (+0.3)**	**98.8 (+0.3)**	83.1 (+0.7)	**70.6 (+0.1)**	37.3 (-0.4)

in some cases. The addition of the evidence head not only does not have a negative impact on the performance of the model, it even further improves the model performance. The evidence head is essentially a classification head with excellent performance and has been used to completely replace the classification head in some studies. Additional evidence-based learning of classification information improves model performance. Because the evidence head only replaces the softmax layer of the classification head with a nonnegative activation layer in the network implementation, the calculation of the target detection network does not increase significantly during inference.

The addition of contrastive learning further comprehensively improves the performance of the model. We observe that for the AP50 metric, the performance improvement of the evidence head is sometimes higher than that of contrastive learning. However, not all models and datasets exhibit this phenomenon. For the other two AP indicators that reflect the overall performance of the model [AP75 and AP (0.5:0.95)] and the $F1$ score that reflects the single-point performance of the model, contrastive learning has a significant performance improvement compared to evidence head. As for the recall and precision, it can be observed in Tables II–V that the model performance improvement is mainly reflected in the precision. First, the epistemic uncertainty comes from the classification branches, making them naturally more sensitive to classification information. Second, regardless of whether the three contrastive learning modules bring the intraclass distance closer or the interclass distance farther, they are all strengthening the classification boundary between positive and negative samples, which improves the precision performance more directly. The improvement in recall is mainly due to the contrastive learning between positive samples tightening the intraclass features.

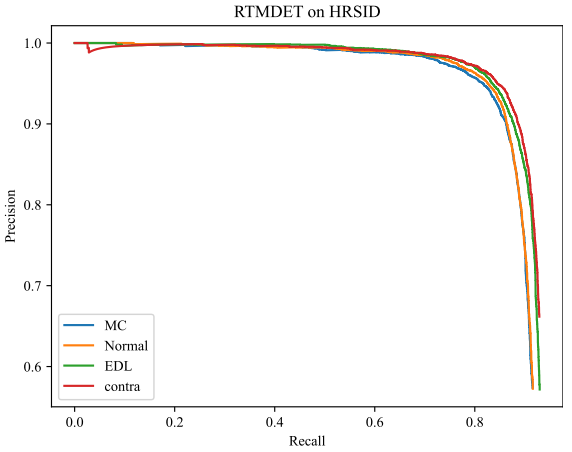
It can be observed from the FPS indicators in Tables II–VII that adding evidence headers only causes a slight decrease in inference speed. The main reason is that the evidence header only adds a set of evidence predictions to the classification branch of the object detection model. In fact, for large orders of magnitude, the variation of parameter quantity can be basically ignored.

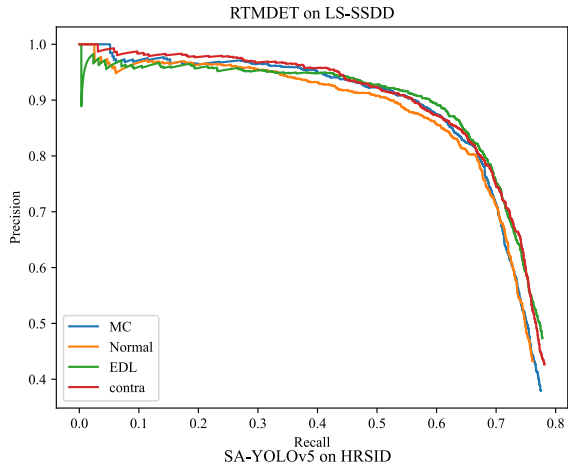
Figs. 5 and 6 shows the P-R curves of RTMDET and SA-YOLOv5 models after different processing on HRSID and LS-SSDD datasets. It can be seen from the figure that the performance of the network model after adding the evidence head is significantly improved. Since the P-R curve in Fig. 5 is based on AP50, there is little difference between the network adding the evidence head alone and the network adding the contrastive learning module continuously. The advantages of the contrastive learning module are mainly reflected in the optimal precision and recall of a single point. The P-R curve under AP75 more clearly shows the performance advantages brought by feature contrast. The distinguishability of features is more evident under strict standards.

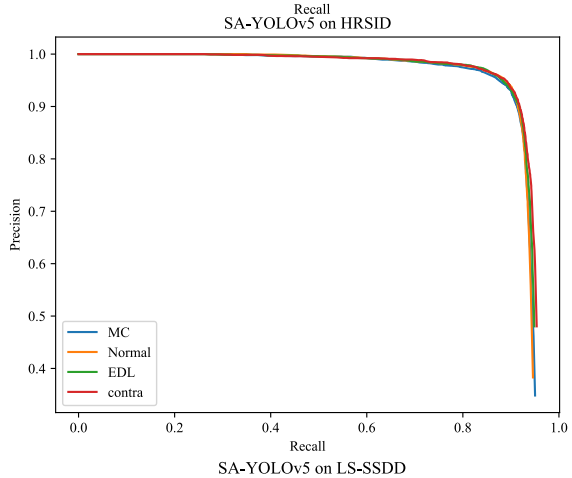
Fig. 7 shows the performance of the RTMDET model on the HRSID dataset, with the left column displaying high-resolution large inshore ships and the right column displaying low-resolution small inshore ships. From the figure, it can be seen that MC dropout may lead to a certain degree of false alarms and missed detections. The addition of evidence heads mainly increases the recall of the model, but also brings some false alarms. Contrastive learning corrected the classification errors of the model, and the false alarm of inshore ships significantly decreased.

The last thing worth showing is to compare the uncertainty changes before and after contrastive learning. As shown in Fig. 8, after adding the contrastive learning module, the uncertainty of the ship target is significantly reduced. It indicates that after sample comparison, the model is more confident in its predictions. It can be seen that the classification scores (confidence) of the predictions mostly also increased, but not as significantly as the uncertainty. Fig. 9 shows the distribution of the uncertainty of all samples with respect to the classification score (confidence). A phenomenon similar to Fig. 8 can be observed: after the addition of the contrastive learning module, the uncertainty of the model prediction decreases collectively, indicating that the model is more confident in its own prediction.

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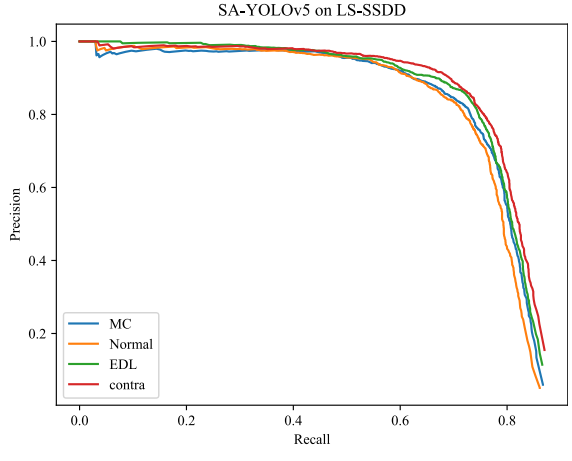
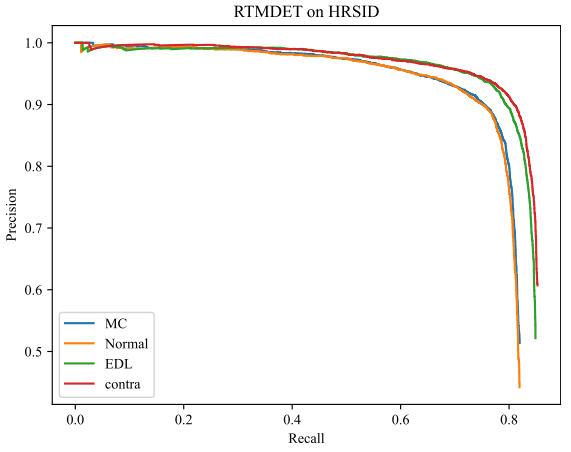
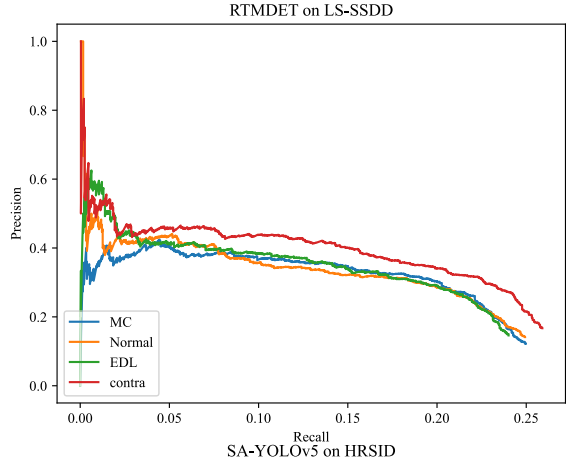
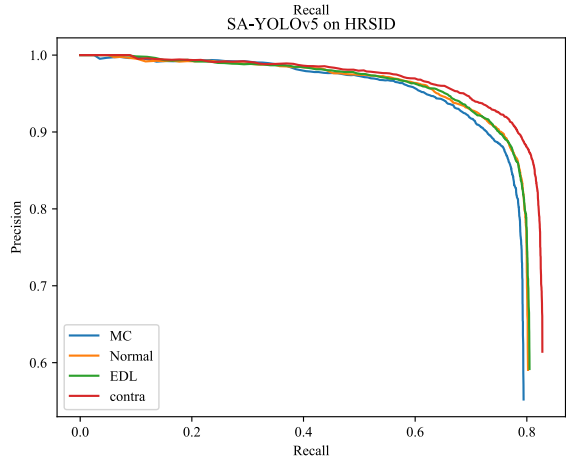


Fig. 5. P-R curve of RTMDET and SA-YOLOv5 (AP50).







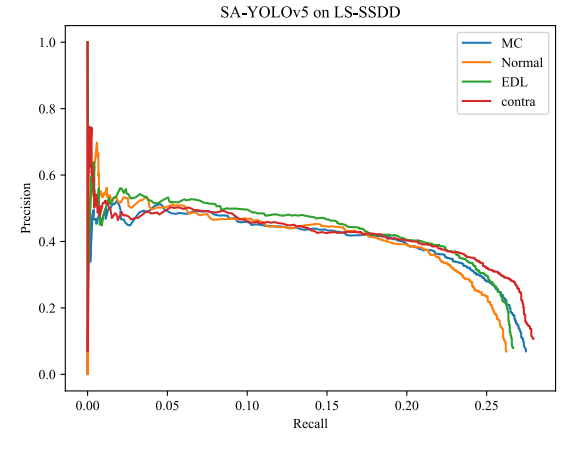


Fig. 6. P-R curve of RTMDET and SA-YOLOv5 (AP75).

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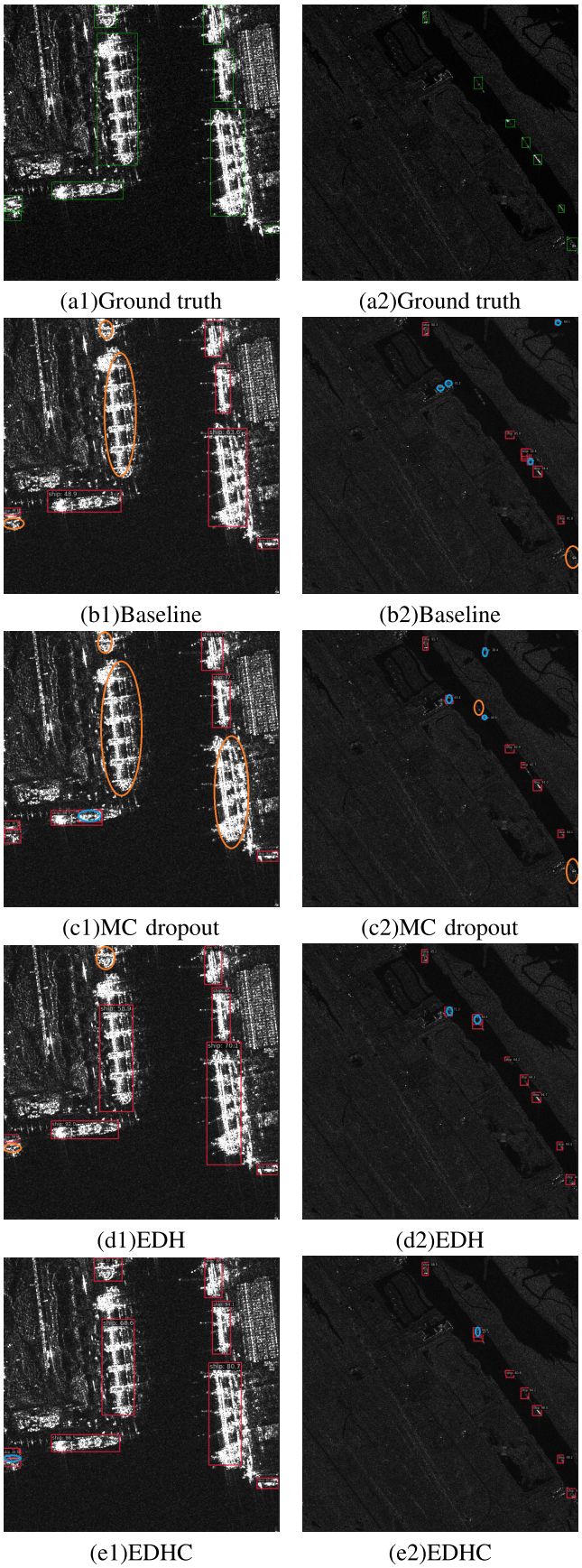


Fig. 7. Performance of the RTMDET model on the HRSID dataset (Green: ground truth, Red: detection results, Blue: false alarm, Orange: missed detection). (a1) and (a2) Ground truth. (b1) and (b2) Baseline. (c1) and (c2) MC dropout. (d1) and (d2) EDH. (e1) and (e2) EDHC.

TABLE VIII
ABLATION EXPERIMENT OF THE CONTRASTIVE LEARNING MODULE

HLTP	LLTP	LTFP	P(%)	R(%)	AP50(%)	AP75(%)	AP(0.5:0.95)(%)
-	-	-	85.9	72.1	79.5	31	15.3
•	-	-	86.0	72.2	79.7	30.8	15.6
-	•	-	85.1	72.8	79.5	30.9	15.6
-	-	•	**86.2**	72.1	79.7	30.9	15.7
•	•	-	86.1	72.4	79.9	31.1	15.5
•	•	•	85.8	**72.9**	**80.9**	**32.2**	**16**

TABLE IX
COMPARISON WITH OTHER METHODS ON THE HRSID DATASET

Method	P(%)	R(%)	AP(%)	FPS
Faster R-CNN	85.0	81.5	80.3	35.6
Cascade R-CNN	86.1	82.3	81.1	27.8
RetinaNet	42.2	81.4	67.9	**38.1**
Double-head R-CNN	87.5	81.9	80.7	16.3
FCOS	77.6	90.2	87.7	35.5
FASC-Net	88.3	80.2	88.3	/
BL-Net	89.7	**91.6**	88.7	/
SA-YOLOv5	93.9	90.0	94.0	37.7
Our(SA-YOLOv5 based)	**94.8**	89.2	**94.8**	37.3

E. Ablation Experiment of Contrastive Learning Module

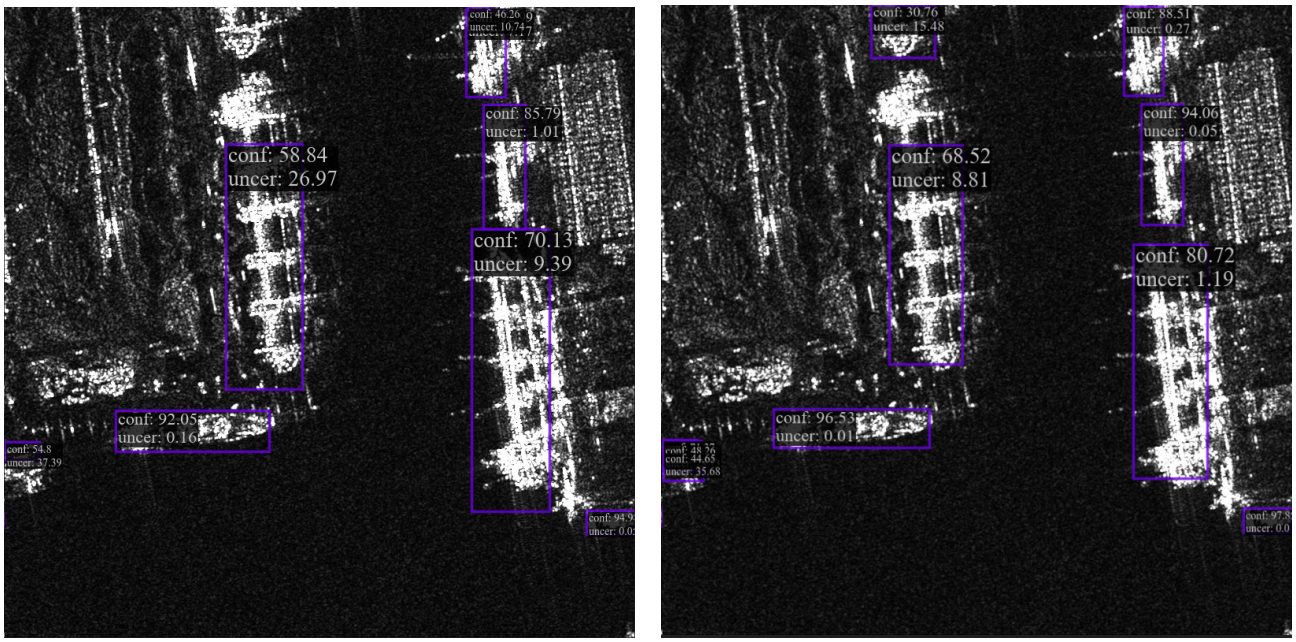
We divided the samples into three groups in the contrastive learning module. To verify and explain the rationality of the existence of the three groups, the following ablation experiments were conducted. The ablation experiment was conducted on the LS-SSDD dataset using the SA-YOLOv5 model. The training steps of the ablation experiment are consistent with the second stage of training described previously.

It can be seen from Table VIII that the model performance is beneficial when the three sets of contrastive learning modules are used separately. HTP samples are actually positive samples with difficult feature extraction or incomplete feature extraction. Shortening the feature distance between them and well-learned LTP samples is conducive to shortening the intraclass distance of positive samples, thereby improving the feature learning of the HTP samples. As a result, various performance metrics have seen a slight improvement. The LTP sample itself is already a sample with good feature learning. On the one hand, the contrastive learning between them can shorten the intraclass distance, but it also causes a certain degree of feature richness reduction. So, we observed a relatively obvious increase in recall, which also caused a decrease in precision. This is worthwhile because the other two sets of contrastive learning modules do not have a clear recall growth function. To illustrate this point, an experiment in which HLTP and LLTP work simultaneously was added. It can be seen that the precision has also been improved on the premise that the recall has increased. The confusion between positive and negative samples was reduced by comparing LTP with FP, which directly contributed to the reduction of false alarms. At the same time, the discrimination of the ship features of positive samples was enhanced. The performance of LTFP was similar to that of HLTP.

F. Comparative Experiment

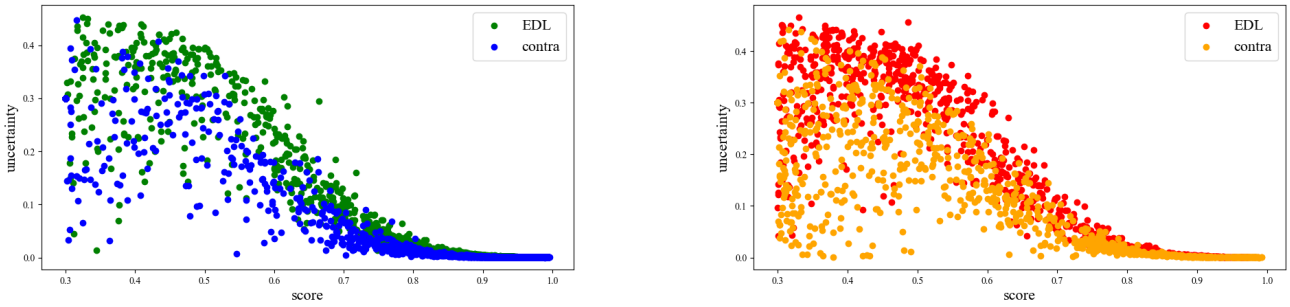
The proposed method is compared with some popular target detection methods, including advanced SAR ship detection methods, as shown in Tables IX–XI. Since many methods do

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(a) (b)

Fig. 8. Uncertainty and confidence before and after contrastive learning (RTMDET on HRSID). (a) EDL. (b) Contrastive learning.



(a) (b)

Fig. 9. Uncertainty change of before and after adding the contrastive learning module (RTMDET on HRSID). (a) TP samples. (b) FP samples.

TABLE X
COMPARISON WITH OTHER METHODS ON THE LS-SSDD DATASET

Method	P(%)	R(%)	AP(%)	FPS
Faster R-CNN	73.7	77.7	74.8	35.6
Cascade R-CNN	84.1	72.7	70.9	27.8
RetinaNet	94.6	55.5	54.3	**38.1**
Double-head R-CNN	74.8	77.8	75.4	16.3
FCOS	73.2	**80.2**	76.0	35.5
L-YOLO	84.8	64.0	73.0	/
SII-Net	68.2	79.3	76.1	/
SA-YOLOv5	83.1	70.5	77.1	37.7
Our(SA-YOLOv5 based)	**85.8**	72.9	**80.9**	37.3

TABLE XI
COMPARISON WITH OTHER METHODS ON THE SSDD DATASET

Method	P(%)	R(%)	AP(%)	FPS
Faster R-CNN [20]	85.4	90.1	89.2	35.6
Cascade R-CNN [43]	90.8	94.1	90.5	27.8
RetinaNet [44]	85.1	91.6	88.9	**38.1**
Double-head R-CNN [45]	91.9	87.9	91.1	16.3
FCOS [46]	84.7	95.8	95.8	35.5
Quad-FPN [47]	95.8	89.5	95.3	/
HR-SDNet [48]	90.1	96.5	90.8	/
SA-YOLOv5 [8]	96.5	97.1	98.5	37.7
Our(SA-YOLOv5 based)	**96.7**	**97.4**	**98.8**	37.3

not provide AP75 metric and AP (0.5:0.95) metric, the AP in these three tables specifically refers to AP50.

The method proposed can achieve SOTA performance on the HRSID, SSDD, and LS-SSDD datasets. On the one hand, this relies on the good performance of the base model SA-YOLOv5. On the other hand, contrastive learning based on epistemic uncertainty provides interpretable performance improvements, which is especially evident in the LS-SSDD dataset. The AP of SA-YOLOv5 based on the proposed method increased by 3.8%, thus achieving SOTA performance on the LS-SSDD dataset.

G. Robustness and Generalization Experiments

To verify the detection performance of our method in large and complex scenes, we select five large-scene images from the LS-SSDD test set. The resolution of all five images

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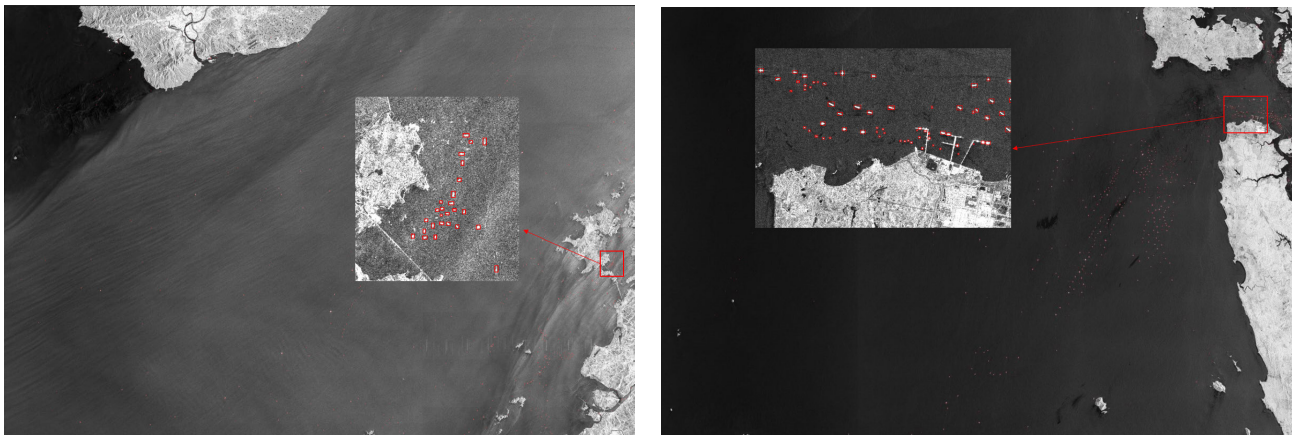


Fig. 10. Large-scale areas (24 000 × 16 000) detection results.

TABLE XII

MODELS TRAINED ON THE HRSID DATASET FOR VALIDATION ON THE SSDD DATASET AND THE LS-SSDD DATASET. (H→S MEANS TRAINED ON HRSID AND VALIDATED ON SSDD, H→L MEANS TRAINED ON HRSID AND VALIDATED ON LS-SSDD)

Method	P(%)		R(%)		F1		AP(%)	
Target	H→S	H→L	H→S	H→L	H→S	H→L	H→S	H→L
Baseline	82.4	40.5	62.5	20.6	71.1	27.3	70.1	17.4
EDHC	**84.5(+2.1)**	**41.7(+1.2)**	**65.2(+2.7)**	**21.8(+1.2)**	**73.6(+2.5)**	**28.6(+1.3)**	**73.3(+3.2)**	**19.5(+2.1)**

was 24 000 × 16 000. The final AP is 74.13% on baseline SA-YOLOv5 and 76.41% on the proposed method. Large-scale scenes showed a slight but acceptable decrease in performance compared to slicing detection. There are some detection results shown in Fig. 10.

To verify the generalization performance of the proposed method. We select models trained on the HRSID dataset for validation on the SSDD dataset and the LS-SSDD dataset. The model is SA-YOLOv5-based. The experimental results are shown in Table XII. Experimental results prove that the method proposed in this article is better than the baseline model in cross-domain situations. We believe that the main reason is feature constraints targeting intraclass imbalance encourage the model to learn more general ship features, ultimately achieving better generalization performance.

V. CONCLUSION

In this article, we propose that SAR ship detection faces the problem of difficult samples under intraclass imbalance. The target detection model outputs interpretable epistemic uncertainty through evidence learning while making predictions. Uncertainties provide intraclass discrimination labels for difficult samples. Contrastive learning is used to improve detection performance. The proposed method is proved to be effective on multiple target detection models and achieves SOTA performance on the HRSID dataset, the SSDD dataset, and the LS-SSDD dataset.

The focus of future related work is to optimize the model with uncertainty in real-time. The current contrastive learning is based on two-stage model training, and the easy-to-use uncertainty optimization should play a role in each iteration of the model. The introduction of aleatoric uncertainty into SAR ship detection is also worthy of further study.

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